Lecture Notes on

Exoplanets

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These notes summarize the fourth Astrophysics Club meeting, where we explored exoplanets, detection methods, and habitability.

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1 What is an Exoplanet?

An exoplanet (extrasolar planet) is any planet orbiting a star outside our Solar System.

Why this matters:

- Tests whether our Solar System is typical.
- Helps refine models of planet formation.
- Allows search for potentially habitable worlds.

2 History of Discovery

- 1992 — First exoplanets discovered orbiting a pulsar.
- 1995 — 51 Pegasi b discovered (first around Sun-like star).

51 Pegasi b:

- A "hot Jupiter."
- Massive gas giant orbiting extremely close to its star.
- Challenged traditional planet formation models.

3 Detection Methods

Finding exoplanets is difficult:

- Planets are small.
- Stars are extremely bright.
- Most planets cannot be directly seen.

3.1 1. Transit Method

When a planet passes in front of its star:

- Star brightness dips slightly.
- This dip is measurable.

Transit depth:

$$\frac{\Delta F}{F} \approx \left(\frac{R_p}{R_*}\right)^2.$$

Thus:

- Transit depth $\rightarrow$ planet radius.

- Time between dips → orbital period.

Used by:

- Kepler Mission
- TESS

3.2 2. Radial Velocity (Doppler Wobble)

Star and planet orbit a shared barycenter.
Star's motion produces Doppler shifts:

$$\frac{\Delta\lambda}{\lambda} = \frac{v}{c}.$$

Where:

- Blueshift → star moving toward Earth.
- Redshift → star moving away.

Radial velocity amplitude:

$$v_* \propto \frac{M_p}{M_*}.$$

Thus:

- Larger planet → stronger wobble.
- Gives planet's minimum mass.

3.3 3. Direct Imaging

- Block star's light.
- Detect faint planet glow.
- Works best for massive planets far from stars.

Very technically challenging.

3.4 4. Gravitational Microlensing

From general relativity:

- Mass curves spacetime.
- Light follows curvature.

If a star passes in front of background star:

- Background star temporarily brightens.

- Additional bump in light curve → planet present.

Advantages:

- Can detect distant planets.
- Sensitive to low-mass planets.

4 Determining Composition

If we know:

- Radius (from transit),
- Mass (from radial velocity),

we compute density:

$$\rho = \frac{M}{\frac{4}{3}\pi R^3}.$$

Interpretation:

- High density → rocky.
- Low density → gaseous.
- Intermediate → water/ice-rich.

4.1 Spectroscopy

Transmission spectroscopy:

- During transit, starlight passes through atmosphere.
- Certain wavelengths absorbed by molecules.

Common atmospheric signatures:

- H_2O
- CO_2
- CH_4
- O_2
- O_3

Emission/reflection spectroscopy:

- Measure light during secondary eclipse.
- Reveals temperature and atmospheric composition.

5 Habitability

A planet is potentially habitable if it can support liquid water.
Key factors:

- Distance from star (habitable zone).
- Atmosphere (greenhouse regulation).
- Planet mass (retain atmosphere).
- Magnetic field (radiation shielding).
- Stable host star.
- Essential chemistry (C, H, O, N).

Habitable zone distance scales roughly as:

$$r \propto \sqrt{L_*}.$$

Where L_* is stellar luminosity.

6 Biosignatures

Signs of life may include:

- Oxygen (O_2)
- Ozone (O_3)
- Methane (CH_4)
- Water vapor (H_2O)

Especially important:

- O_2 and CH_4 coexisting.

Such combinations are chemically unstable without biological replenishment.
Major instruments:

- JWST
- TESS
- Future missions (HabEx, LUVOIR concepts)

7 Promising Systems

- TRAPPIST-1 (7 Earth-sized planets).
- Kepler-452b (super-Earth candidate).
- LHS 1140 b (rocky planet in habitable zone).

These systems guide future observation priorities.

8 Big Picture

Exoplanet science allows us to:

- Understand planetary system diversity.
- Test formation models.
- Search for life beyond Earth.

We are likely living during the first era in human history where detecting life elsewhere is scientifically possible.